

Unit 2: Basic Concepts of Hadoop

2.1 Hadoop Ecosystem Overview

The Hadoop Ecosystem is a platform or framework that provides various services to solve big data problems. It includes core components for storage and processing, along with "top-level" projects for specialized tasks.

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- i. **Core Layers:** HDFS (Storage), YARN (Resource Management), MapReduce/Spark (Processing).
- ii. **Data Access:** Hive (SQL), Pig (Scripting).
- iii. **Data Integration:** Sqoop (RDBMS to Hadoop), Flume (Logs to Hadoop).
- iv. **Coordination & Management:** ZooKeeper (Coordination), Ambari (Monitoring).

2.2 Hadoop Distributed File System (HDFS)

HDFS is a **distributed, scalable, and fault-tolerant** file system designed to run on commodity hardware.

Key Functionality & Architecture

1. Master-Slave Architecture:

- **NameNode (Master):** The "Brain." It manages the file system namespace and metadata (file names, permissions, and location of blocks). It does NOT store the actual data.
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- **DataNodes (Slaves):** The "Muscle." They store the actual data blocks and perform read/write operations as directed by the NameNode.
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2. Data Replication:

- HDFS splits files into **Blocks** (default 128MB).
- Each block is replicated (usually 3 times) across different nodes. This ensures **Fault Tolerance**—if one node fails, the data is safe on others.
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3. **Rack Awareness:** HDFS is smart enough to store replicas on different physical racks to protect against a total rack power failure.

2.3 MapReduce Framework

MapReduce is the software framework for writing applications that process vast amounts of data in parallel on large clusters.

The Workflow Logic

1. **Map Phase:** The input data is split into independent chunks. The mapper function processes these chunks to produce intermediate **Key-Value pairs** (K, V).
2. **Shuffle and Sort:** The framework gathers all values associated with the same key and sorts them. This is the "bridge" between Map and Reduce.
3. **Reduce Phase:** The reducer function takes the sorted (K, list of values) and aggregates them into a final result.

Simple MapReduce Implementation (Python/Streaming)

For the exam, you can use this simple logic for a **Word Count** program:

```
# mapper.py
```

```
import sys
```

```
for line in sys.stdin:
```

```
    words = line.strip().split()
```

```
    for word in words:
```

```
        # Output: word 1
```

```
        print(f'{word}\t1')
```

```
# reducer.py
```

```
import sys
```

```
current_word = None
```

```
current_count = 0
```

```
for line in sys.stdin:
```

```
    word, count = line.strip().split("\t")
```

```
if word == current_word:
    current_count += int(count)
else:
    if current_word: print(f'{current_word}\t{current_count}')
    current_word = word
    current_count = 1
print(f'{current_word}\t{current_count}')
```

2.4 Hadoop YARN (Resource Management)

YARN (Yet Another Resource Negotiator) is the "Operating System" of Hadoop. It separates resource management from data processing.

Main Components

- **ResourceManager (RM):** The global authority. It has two parts:
 - **Scheduler:** Allocates resources (CPU, Memory) to various running applications.
 - **ApplicationsManager:** Accepts job submissions and starts the first container for the ApplicationMaster.
- **NodeManager (NM):** Per-machine agent. It monitors resource usage (containers) on a specific node and reports back to the RM.
- **ApplicationMaster (AM):** One per application. It negotiates resources from the RM and works with the NM to execute and monitor tasks.
- **Container:** A physical collection of resources (RAM, CPU) on a single node where a task runs.

2.5 Hadoop Applications: Hive, Spark, Sqoop, Pig

Application	Technology	Primary Function	Best Use Case
Apache Hive	SQL-on-Hadoop	Data warehousing and ad-hoc querying using HiveQL .	Analysts who know SQL but not Java.
Apache Spark	In-Memory Engine	Fast, general-purpose data processing (100x faster than MapReduce).	Real-time streaming and Machine Learning.
Apache Sqoop	Data Transfer	SQL-to-Hadoop . Efficiently transfers data between RDBMS (MySQL, Oracle) and HDFS.	Moving legacy business data into Big Data clusters.
Apache Pig	Scripting Platform	High-level platform for creating execution plans using Pig Latin .	Complex ETL data pipelines and transformations.